

FAM — Technical Specification Sheet

Project Overview

FAM (Family Event & Hierarchy System) is a scalable wedding family tree platform that combines:

- Dynamic family tree visualization
- Event access & management
- QR-based guest onboarding
- AI-powered moderation & recommendations
- Social networking for event guests
- Multi-role administration system

1. SYSTEM ARCHITECTURE

Architecture Type

- **Microservices**
- REST API-based communication
- Mobile-first Progressive Web App (PWA)

2. TECH STACK

Frontend Stack

Technology	Purpose
React.js	Frontend Framework
Tailwind CSS	Styling/UI
React Router DOM	Routing
Axios	API Communication
Framer Motion	Animations
Mermaid.js	Family Tree Rendering
Vite	Build Tool

React Query / TanStack Query	API Caching
PWA Support	Installable App
Socket.IO Client	Real-time Chat/Notifications
React Hook Form	Form Handling
QR Scanner Library	QR Event Joining

Backend Stack

Technology	Purpose
Node.js	Runtime
Express.js	Backend Framework
MongoDB	Main Database
Mongoose	ODM
JWT	Authentication
Nodemailer	Email OTP
Socket.IO	Real-time Communication
Multer	File Upload
Cloudinary	Media Storage
CORS	Cross-Origin Security
Helmet.js	Security Headers
Morgan	Logging
Bcrypt	Encryption
Rate Limiter	API Protection
PGVector (Experimental)	AI Relationship Mapping
Railway/Render	Deployment
PM2	Process Management

3. DATABASE DESIGN

Major Collections

Collection	Purpose
users	User accounts
events	Wedding/event data
organizers	Organizer permissions
familyTrees	Tree structure
relationships	Parent/partner relations
chats	Event chat messages
notifications	Alerts/reminders
logs	Audit trails
reports	User reports
schedules	Event timeline
venues	Venue maps/layouts
dietaryPreferences	Allergy/food data
refunds	Refund requests
mediaUploads	Uploaded photos/videos
aiReports	Wrong AI response reports

4. AUTHENTICATION SYSTEM

Authentication Type

- Email OTP Authentication
- Passwordless Login
- JWT Token-Based Sessions
- Refresh Token Support

Security Features

- Rate Limiting
 - Device Validation
 - Session Expiry
 - Soft Suspension
 - Role-Based Access Control (RBAC)
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5. USER ROLES

Role	Permissions
Admin	Full system access
Organizer	Moderate/manage assigned event
User/Family Member	Standard participation

6. FRONTEND PAGE ESTIMATION

Public Pages

Page	Count
Landing Page	1
Login/OTP	2
Event Join Page	1
QR Scan Page	1
Event Discovery	1
Theme Preview	1

Public Total: 6 Pages

User Pages

Page	Count
Dashboard	1
Event Home	1
Family Tree	1
Profile	1
Notifications	1
Chat System	1
Guest Discovery	1
Schedule Timeline	1
Venue Visualization	1
Contact Exchange	1
Dietary Preferences	1
Photo Gallery	1
Pull My Photos	1
Settings	1

User Total: 14 Pages

Organizer Pages

Page	Count
User Moderation	1
Schedule Management	1
Media Approval	1
Reports Panel	1
Event Controls	1

Organizer Total: 5 Pages

Admin Pages

Page	Count
Admin Dashboard	1
Event Management	1
Global Logs	1
User Management	1
AI Reports	1
Analytics	1
System Monitoring	1

Admin Total: 7 Pages

TOTAL FRONTEND PAGES

≈ 32–42 Pages

(Depending on modal-based or dedicated-page implementation)

7. API ESTIMATION

Authentication APIs

- Register/Login
- OTP Send
- OTP Verify
- Logout
- Refresh Token
- Session Validation

≈ 6 APIs

User APIs

- Profile CRUD
- Notification APIs
- Contact APIs
- Settings APIs

≈ 15 APIs

Event APIs

- Create Event
- Join Event
- Update Event
- Delete Event
- Access Validation
- QR Validation
- Invitation APIs

≈ 20 APIs

Family Tree APIs

- Add Relation
- Update Relation
- Remove Relation
- Generate Tree
- Tree Validation
- Hierarchy Validation

≈ 18 APIs

Chat APIs

- Send Message
- Report Message
- Fetch Messages

≈ 10 APIs

Media APIs

- Upload Media
- Moderate Media
- NSFW Validation
- Fetch Gallery
- Pull Photos

≈ 15 APIs

Organizer/Admin APIs

- User Moderation
- Logs
- Analytics
- Suspension
- Reports
- Refunds

≈ 25 APIs

AI/Recommendation APIs

- Tourist Recommendations
- Nearby Events
- AI Reporting
- Interest Matching

≈ 10 APIs

TOTAL API COUNT

≈ 110–140 REST APIs

8. CORE SYSTEM MODULES

Module	Complexity
Authentication	Medium
Event Management	High
Event Scheduling	Moderate
Family Tree Engine	Very High
QR System	Medium
AI Recommendation Engine	High
Real-Time Chat	High
Media Moderation	High
Venue Visualization	Medium
Notification System	High
Audit Logging	High

9. FAMILY TREE ENGINE

Features

- Dynamic relation insertion
- Horizontal visualization
- Step relations
- Multiple marriages
- Same-gender relations
- Pets/friends support
- Deceased member support

Algorithms

Purpose	Algorithm
Generation Detection	BFS
Hierarchy Creation	DAG/Tree Traversal
Conflict Detection	Graph Validation
Duplicate Detection	Hashing

Age Validation	Rule Engine
Dynamic Expansion	Recursive Traversal

10. REAL-TIME FEATURES

Feature	Technology
Chat	Socket.IO
Live Notifications	WebSockets
Schedule Updates	Event Emitters

11. SCALABILITY PLAN

Initial Capacity

- 5,000 Concurrent Users

Target Scalability

- 1M+ Users

Scaling Features

- Auto Scaling
 - Horizontal Scaling
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12. SECURITY FEATURES

Feature	Purpose
JWT Authentication	Secure sessions
RBAC	Permission isolation
Rate Limiting	API abuse prevention
Helmet.js	HTTP security

Input Validation	Injection prevention
Soft Delete	Recovery support
Audit Logs	Accountability
NSFW Detection	Unsafe media filtering
Spam Detection	Abuse prevention

13. STORAGE ARCHITECTURE

Media Strategy

- External storage preferred
- Cloudinary / Google Photos / iCloud integration

Benefits

- Reduced DB load
 - Faster delivery
 - Better scalability
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14. AI FEATURES

Feature	Purpose
Face Recognition	Photo tagging
Pull My Photos	User-specific media fetch
AI Recommendations	Tourist spots/restaurants
Offensive Pattern Detection	Chat moderation
NSFW Detection	Media/Word moderation
AI Reporting	Wrong answer reporting

15. MAJOR EDGE CASE SYSTEMS

Category	Example
Security	QR sharing
Moderation	False NSFW detection
Scalability	Sudden viral traffic
Hierarchy	Impossible family relations
Scheduling	Mid-event changes
Permissions	Privilege escalation
Refunds	Policy disputes
Logs	Massive audit storage

16. DEPLOYMENT PLAN

Layer	Platform
Frontend	Vercel
Backend APIs	Railway/Render
Database	MongoDB Atlas
Media Storage	Cloudinary

17. FUTURE EXPANSION

- Native Mobile Apps
 - AI Family Story Generator
 - AI Wedding Assistant
 - Smart Seating Arrangement
 - Blockchain-Based Invitations
 - AR Venue Navigation
 - AI Relationship Suggestions
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18. PROJECT COMPLEXITY

Area	Complexity
Frontend	High
Backend	Very High
Database	High
AI Features	Very High
Scalability	Enterprise-Level
Family Tree Logic	Complex Graph System

FINAL ESTIMATION

Metric	Estimated Value
Frontend Pages	32–42
REST APIs	110–140
Database Collections	15+
Major Modules	12+
Real-Time Systems	4+
Concurrent User Goal	1M+